

AKASH KUNDU

B.Tech in Computer Science & Engineering · Techno India University, Kolkata
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Summary

Computer Science undergraduate specializing in Machine Learning, Computer Vision, and Artificial Intelligence. Built a foundational ML skill set through a Google for Developers-supported AICTE internship. Bridge AI research and software engineering by developing full-stack applications, containerizing predictive models via Docker, and applying Microsoft Azure infrastructure. Engineered solutions embedding locally-deployed, GPU-accelerated models, achieving sub-50ms inference latency. Aiming to drive AI engineering initiatives by accelerating end-to-end model pipelines and collaborating in cross-functional teams.

Education

Techno India University <i>B.Tech — Computer Science & Engineering</i> CGPA: 7.61/10 (through 5th Semester)	08/2023 – 05/2026 Kolkata
Arambagh Vivekananda Academy <i>Higher Secondary — Science (Class XII)</i> Percentage: 71.6%	04/2021 – 05/2023 Arambagh

Experience

AI-ML Virtual Internship (Supported by Google for Developers) <i>AICTE & EduSkills</i>	10/2025 – 12/2025 Remote
- Completed a 10-week curriculum mastering foundational Artificial Intelligence and Machine Learning concepts. - Applied core ML frameworks and data processing techniques, establishing the technical groundwork for advanced independent projects.	

Projects

AI-based Eye Disease Predictor [Completed]  <i>Python · PyTorch · OpenCV</i> 01/2026 – 04/2026	- Designed an end-to-end computer vision pipeline using PyTorch, fine-tuning deep Convolutional Neural Networks (CNNs), specifically EfficientNet, to hierarchically classify 7 ocular diseases with >85% accuracy. - Constructed an automated data preprocessing pipeline utilizing Pandas to ingest, clean, and augment 5,000+ ocular images before model training. - Accelerated model workflows by transitioning from CPU to performance GPU environments, cutting epoch training time by 70%. - Automated PDF diagnostics with Grad-CAM, slashing manual review time by 40% for clinical interpretability.
COPPER — AI Desktop Productivity Assistant [In Progress]  <i>Tauri · FastAPI · Ollama</i> 03/2026 – Present (Concurrent)	- Developed a secure, local desktop assistant using Tauri, FastAPI, and Ollama, executing large language models 100% offline with <50ms inference latency. - Implemented asynchronous voice-to-text processing pipelines handling 10+ daily queries, integrating a persistent local database memory layer to improve daily workflow efficiency by 30%.
AI Hardware Benchmark Engine [In Progress]  <i>Python · FastAPI · Docker</i> 02/2026 – Present (Concurrent)	- Built a predictive machine learning engine using Python and CatBoost to forecast hardware performance metrics across 3 heavy workloads, achieving 95% prediction reliability. - Formulated an automated data pipeline to scrape and structure specifications from 6+ hardware tables for optimal model training. - Containerized the FastAPI serving layer using Docker to ensure consistent environment deployment, achieving <30ms response times for hardware upgrade recommendations.
E2EE Multi-User Chat Application [Completed]  <i>React · Node.js · Java</i> 11/2025 – 01/2026	- Programmed a high-throughput, multi-threaded Java and Node.js backend supporting 50+ concurrent users with 256-bit E2E encryption and JWT authentication. - Orchestrated WebSocket layers and integrated MongoDB to manage 10+ active chat rooms, utilizing Cloudflare/Ngrok tunneling to achieve sub-100ms message delivery.

Leadership

- Team Lead, Smart India Hackathon (SIH) 2024 & 2025:** Led a 6-member cross-functional team to the second level of India's premier national hackathon for two consecutive years.

Skills

Languages: Python, Java, JavaScript (ES6+), C++, SQL

AI & Machine Learning: PyTorch, Computer Vision (CNNs, EfficientNet), Local LLMs (Ollama), scikit-learn, CatBoost, Pandas

Web & Model Serving: FastAPI, Node.js, React.js, Tailwind CSS, REST APIs, WebSockets

Cloud & Infrastructure: Microsoft Azure, Docker, Kubernetes, CI/CD (GitHub Actions), PostgreSQL, MongoDB, Ngrok

Tools & Soft Skills: Git, GitHub, Cross-functional Teamwork, Technical Communication

Certifications

- Deloitte Data Analytics Simulation** (05/2026) | **AI-ML Virtual Internship — AICTE** (12/2025)
- Deloitte Cyber Simulation** (07/2025) | **Getting Started with Enterprise AI** (08/2024)